

Luke Song

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Hello! I'm Luke.

I have a background in applied machine learning, web design, game development, and robotics. Currently, I'm broadly interested in reinforcement learning and digital simulations of real world phenomena, but I enjoy solving all sorts of problems and the feelings that come with it.

Education

University of California, Los Angeles

Grad. 2030

B.S in Computer Science

- Courses: Introduction to Computer Science II, Introduction to Computer Organization, Logic Design of Digital Systems, Calculus of Several Variables.

Projects

Emergent Colony | C#, Unity 2D

Jan 2026 - May 2026

- Senior research project. Created simulation of ant colony against evolving predators.
- Implemented pheromone-based distributed decision-making, allowing ants to coordinate using only local environmental information and adapt to static predators.
- Designed defensive and offensive behavioral strategies for ants in response to moving predators.
- Modeled predator evolution under changing ant survival strategies.

Reinforcement Learning for Connect Four | Python, Pytorch

Aug 2026

- Developed Reinforcement Learning model for the game of Connect Four
- Built and optimized custom modular two-part (1) modified monte carlo tree search and (2) ResNet-based dual-head policy and value network.

Experience

COSMOS: Brain Inspired Neural Networks

Summer 2024

- Developed task-driven reinforcement learning model to simulate human running under the effects of ethanol.
- Evaluated and compared three reinforcement learning architectures: DQN, DDPG, and TD3.
- Analyzed and linked simulated behavioral changes to real-world effects of alcohol consumption.

Awards

International Logic Olympiad, Champion

May 2025

- Top 5% in scores of all participants, with over 4000 participants.
- 1st place team in finalist round at Stanford (top 50 scorers).

Skills

- **Language & Tools:** Python, PyTorch, Tensorflow, Numpy, Pandas, JavaScript, HTML/CSS.
- **Concepts:** NLP, Transformers, Reinforcement Learning, Deep Learning.